Clipping versus coiling for aneurysmal subarachnoid hemorrhage: a systematic review and meta-analysis of prospective studies

Journal name: Neurosurgical review

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Supplemental Table 1: Search strategy.

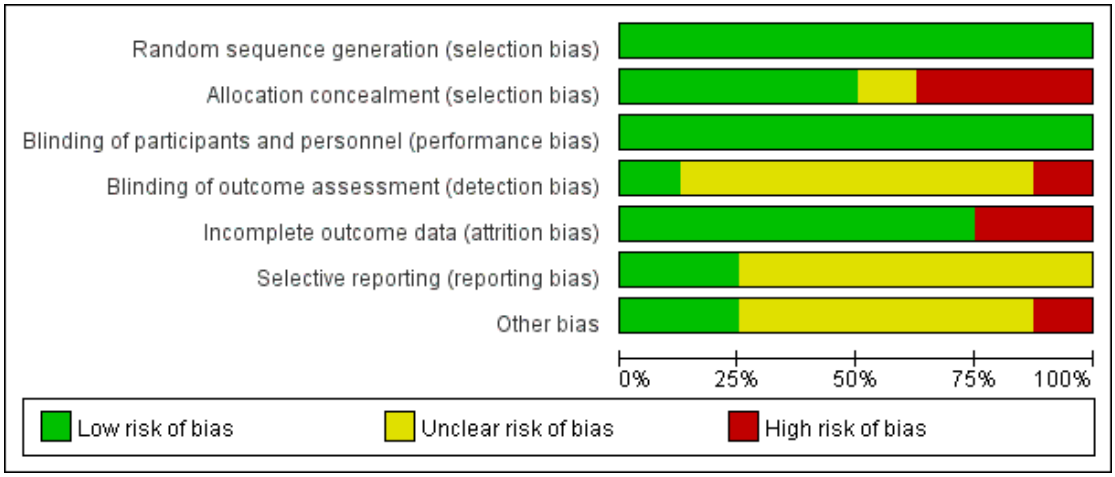
Search set	Embase	Items found
#1	exp intracranial aneurysm/	
#2	exp brain artery aneurysm/	
#3	(intracranial or cerebral or Brain or "Basilar Artery" or Saccular or "Anterior Communicating Artery" or "Posterior Communicating Artery" or "Middle Cerebral Artery" or "Posterior Cerebral Artery" or "Posterior Cerebral Artery" or "Anterior circulation" or "Posterior circulation" or Berry or subarachnoid).ab,kw,ti.	
#4	(aneurysm* or hemorrhage*).ab,kw,ti.	
#5	#3 and #4	
#6	exp subarachnoid hemorrhage/	
#7	#1 or #2 or #5 or #6	
#8	exp craniotomy/	
#9	exp craniectomy/	
#10	exp aneurysm clip/	
#11	exp microsurgery/	
#12	(Craniotom* or Craniectom* or clip* or microsurg*).ab,kw,ti.	
#13	#8 or #9 or #10 or #11 or #12	
#14	exp coil embolization/	
#15	exp endovascular aneurysm repair/ or exp endovascular surgery/	
#16	exp neuroradiology/	
#17	exp prosthesis/	
#18	exp prosthesis/	
#19	exp implant/	
#20	exp endoprosthesis/	
#21	(coil* or emboli* or Embolotherap* or endovascular or Intravascular or neuroradiology or Guglielmi or Prosthes* or Implant* or Endoprosthes*).ab,kw,ti.	
#22	#14 or #15 or #16 or #17 or #18 or #19 or #20 or #21	
#23	exp death/	
#24	exp morbidity/	
#25	exp mortality/	
#26	exp fatality/	
#27	exp dependent personality disorder/	
#28	exp disability/	
#29	exp treatment outcome/ or exp clinical outcome/ or exp Glasgow outcome scale/	
#30	exp comparative effectiveness/	
#31	exp prognosis/	
#32	exp cerebrovascular accident/ or exp stroke/ or exp brain ischemia/ or exp Rankin scale/	
#33	exp complication/	
#34	exp brain vasospasm/	
#35	exp infarction/	
#36	exp brain infection/	
#37	exp hydrocephalus/	
#38	exp recanalization/	
#39	exp hospitalization/	

#40	(Death or Mortalit* or Fatalit* or Morbidit* or Dependenc* or Disabl* or outcome or Efficacy or Effectiveness or Prognos* or "Glasgow Outcome Scale" or gos or "Modified Rankin Score" or mrs or Complication* or Vasospasm* or Angiospasm* or "Vascular Spasm*" or "Cerebrovascular Spasm*" or Ischemi* or Infarct* or Stroke* or "Intracranial infection" or Hydrocephal*).ab,kw,ti.	
#41	(Rebleed or Obliteration* or Occlusion* or Recanalisation or "Hospital stay*" or "Length of Stay" or "Stay Length*" or (Neurologic* and (deficit* or Symptom* or Sign or Dysfunction* or Manifestation*))).ab,kw,ti.	
#42	#23 or #24 or #25 or #26 or #27 or #28 or #29 or #30 or #31 or #32 or #33 or #34 or #35 or #36 or #37 or #38 or #39 or #40 or #41	
#43	("Randomized Controlled" or "Randomized Clinical" or "Controlled Clinical" or RCT or Cohort or "Concurrent Stud*" or Prospect*).ab,kw,ti.	
#44	exp randomized controlled trial/	
#45	exp clinical trial/ or exp controlled study/	
#46	exp cohort analysis/	
#47	exp prospective study/	
#48	#43 or #44 or #45 or #46 or #47	
#49	#7 and #13 and #22 and #42 and #48	2352
	Web of science	
#1	TS=((intracranial OR cerebral OR Brain OR "Basilar Artery" OR Saccular OR "Anterior Communicating Artery" OR "Posterior Communicating Artery" OR "Middle Cerebral Artery" OR "Posterior Cerebral Artery" OR "Anterior Cerebral Artery" OR "Anterior circulation" OR "Posterior circulation" OR Berry OR subarachnoid) AND (aneurysm* OR hemorrhage*))	
#2	TS=(Craniotom* OR Craniectom* OR clip* OR microsurg*)	
#3	TS=(coil* OR emboli* OR Embolotherap* OR endovascular OR Intravascular OR neuroradiology OR Guglielmi OR Prosthes$s OR Implant$ OR Endoprosthes$s)	
#4	TS=(Death OR Mortalit* OR Fatalit* OR Morbidit* OR Dependenc* OR Disabl* OR outcome OR Efficacy OR Effectiveness OR Prognos* OR "Glasgow Outcome Scale" OR gos OR "Modified Rankin Score" OR mrs OR Complication* OR Vasospasm$ OR Angiospasm$ OR "Vascular Spasm$" OR "Cerebrovascular Spasm$" OR Ischemi* OR Infarct* OR Stroke$ OR "Intracranial infection" OR Hydrocephal* OR Rebleed OR Obliteration$ OR Occlusion$ OR Recanalisation OR "Hospital stay$" OR "Length of Stay" OR "Stay Length$" OR (Neurologic* AND (deficit$ OR Symptom$ OR Sign$ OR Dysfunction$ OR Manifestation$)))	
#5	TS=("Randomized Controlled" OR "Randomized Clinical" OR "Controlled Clinical" OR RCT OR Cohort OR "Concurrent Stud*" OR Prospect*)	
#6	#5 AND #4 AND #3 AND #2 AND #1 index=SCI-EXPANDED, SSCI	653
	Cochrane Library	
#1	(intracranial OR cerebral OR Brain OR "Basilar Artery" OR Saccular OR "Anterior Communicating Artery" OR "Posterior Communicating Artery" OR "Middle Cerebral Artery" OR "Posterior Cerebral Artery" OR "Anterior Cerebral Artery" OR "Anterior circulation" OR "Posterior circulation" OR Berry OR subarachnoid):ti,ab,kw	
#2	(aneurysm* OR hemorrhage*):ti,ab,kw	
#3	#1 AND #2	
#4	MeSH descriptor: [Intracranial Aneurysm] explode all trees	
#5	MeSH descriptor: [Subarachnoid Hemorrhage] explode all trees	

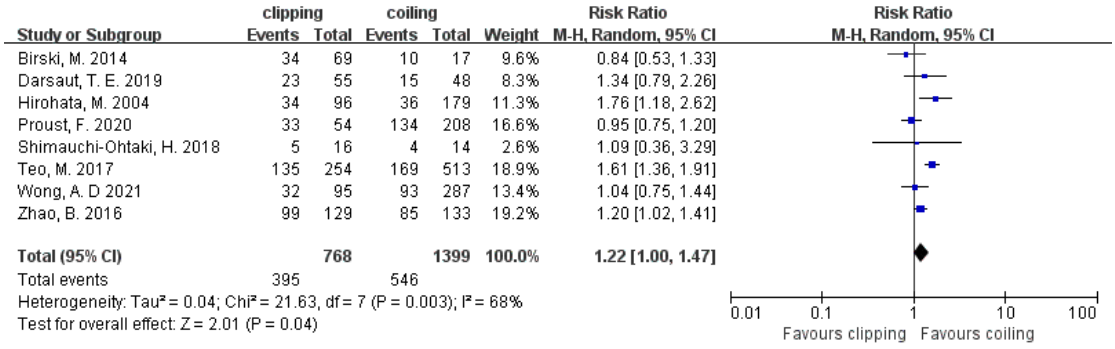
#6	#3 OR #4 OR #5	
#7	(Craniotom* OR Craniectom* OR clip* OR microsurg*):ti,ab,kw	
#8	MeSH descriptor: [Craniotomy] explode all trees	
#9	MeSH descriptor: [Microsurgery] explode all trees	
#10	#7 OR #8 OR #9	
#11	(coil* or emboli* or Embolotherap* or endovascular or Intravascular or neuroradiology or Guglielmi or Prosthes* or Implant* or Endoprosthes*):ti,ab,kw	
#12	MeSH descriptor: [Embolization, Therapeutic] explode all trees	
#13	MeSH descriptor: [Prostheses and Implants] explode all trees	
#14	#11 OR #12 OR #13	
#15	(Death OR Mortalit* OR Fatalit* OR Morbidit* OR Dependenc* OR Disabl* OR outcome OR Efficacy OR Effectiveness OR Prognos* OR "Glasgow Outcome Scale" OR gos OR "Modified Rankin Score" OR mrs OR Complication* OR Vasospasm$ OR Angiospasm$ OR "Vascular Spasm$" OR "Cerebrovascular Spasm$" OR Ischemi* OR Infarct* OR Stroke$ OR "Intracranial infection" OR Hydrocephal* OR Rebleed OR Obliteration$ OR Occlusion$ OR Recanalisation OR "Hospital stay$" OR "Length of Stay" OR "Stay Length$" OR (Neurologic* AND (deficit$ OR Symptom$ OR Sign$ OR Dysfunction$ OR Manifestation$))):ti,ab,kw	
#16	MeSH descriptor: [Death] explode all trees	
#17	MeSH descriptor: [Mortality] explode all trees	
#18	MeSH descriptor: [Morbidity] explode all trees	
#19	MeSH descriptor: [Prognosis] explode all trees	
#20	MeSH descriptor: [Glasgow Outcome Scale] explode all trees	
#21	MeSH descriptor: [Vasospasm, Intracranial] explode all trees	
#22	MeSH descriptor: [Ischemia] explode all trees	
#23	MeSH descriptor: [Neurologic Manifestations] explode all trees	
#24	MeSH descriptor: [Infarction] explode all trees	
#25	MeSH descriptor: [Stroke] explode all trees	
#26	MeSH descriptor: [Hydrocephalus] explode all trees	
#27	MeSH descriptor: [Length of Stay] explode all trees	
#28	#15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27	
#29	("Randomized Controlled" or "Randomized Clinical" or "Controlled Clinical" or RCT or Cohort or "Concurrent Stud*"or Prospect*):ti,ab,kw	
#30	MeSH descriptor: [Randomized Controlled Trial] explode all trees	
#31	MeSH descriptor: [Cohort Studies] explode all trees	
#32	#29 OR #30 OR #31	
#33	#6 AND #10 AND #14 AND #28 AND #32	290
	Pubmed	
#1	Search (((((((((coil*[Title/Abstract]) OR emboli*[Title/Abstract]) OR Embolotherap*[Title/Abstract]) OR endovascular[Title/Abstract]) OR Intravascular[Title/Abstract]) OR neuroradiology[Title/Abstract]) OR Guglielmi[Title/Abstract]) OR Prosthes*[Title/Abstract]) OR Implant*[Title/Abstract]) OR Endoprosthes*[Title/Abstract]	
#2	Search (((((Craniotom*[Title/Abstract]) OR Craniectom*[Title/Abstract]) OR clip*[Title/Abstract]) OR microsurg*[Title/Abstract]))	

#3	Search (((Neurologic*[Title/Abstract]) AND ((((((deficit*[Title/Abstract]) OR Symptom*[Title/Abstract]) OR Sign[Title/Abstract]) OR Signs[Title/Abstract]) OR Dysfunction*[Title/Abstract]) OR Manifestation*[Title/Abstract]))) OR ((((((((((((((((((((((((((((((Death*[Title/Abstract]) OR Mortalit*[Title/Abstract]) OR Fatalit*[Title/Abstract]) OR Morbidit*[Title/Abstract]) OR Dependenc*[Title/Abstract]) OR Disabl*[Title/Abstract]) OR outcome[Title/Abstract]) OR Efficacy[Title/Abstract]) OR Effectiveness[Title/Abstract]) OR Prognos*[Title/Abstract]) OR "Glasgow Outcome Scale"[Title/Abstract]) OR GOS[Title/Abstract]) OR "Modified Rankin Score"[Title/Abstract]) OR mRS[Title/Abstract]) OR Complication*[Title/Abstract]) OR Vasospasm*[Title/Abstract]) OR Angiospasm*[Title/Abstract]) OR "Vascular Spasm"[Title/Abstract]) OR "Vascular Spasms"[Title/Abstract]) OR "Cerebrovascular Spasm"[Title/Abstract]) OR "Cerebrovascular Spasms"[Title/Abstract]) OR Ischemi*[Title/Abstract]) OR Infarct*[Title/Abstract]) OR Stroke*[Title/Abstract]) OR infect*[Title/Abstract]) OR Hydrocephal*[Title/Abstract]) OR Rebleed[Title/Abstract]) OR Obliteration*[Title/Abstract]) OR Occlusion*[Title/Abstract]) OR Recanalisation*[Title/Abstract]) OR "Hospital stay"[Title/Abstract]) OR "Hospital stays"[Title/Abstract]) OR "Length of Stay"[Title/Abstract]) OR "Stay Length"[Title/Abstract]) OR "Stay Lengths"[Title/Abstract])
#4	Search (((aneurysm*[Title/Abstract]) OR hemorrhage*[Title/Abstract])) AND ((((((((((((intracranial[Title/Abstract]) OR cerebral[Title/Abstract]) OR Brain[Title/Abstract]) OR "Basilar Artery"[Title/Abstract]) OR Saccular[Title/Abstract]) OR "Anterior Communicating Artery"[Title/Abstract]) OR "Posterior Communicating Artery"[Title/Abstract]) OR "Middle Cerebral Artery"[Title/Abstract]) OR "Posterior Cerebral Artery"[Title/Abstract]) OR "Anterior Cerebral Artery"[Title/Abstract]) OR "Anterior circulation"[Title/Abstract]) OR "Posterior circulation"[Title/Abstract]) OR Berry[Title/Abstract]) OR subarachnoid[Title/Abstract])
#5	Search ("Intracranial Aneurysm"[Mesh]) OR "Subarachnoid Hemorrhage"[Mesh]
#6	Search ((((((((((((((("Death"[Mesh]) OR "Mortality"[Mesh]) OR "Morbidity"[Mesh]) OR "Outcome Assessment (Health Care)"[Mesh]) OR "Glasgow Outcome Scale"[Mesh]) OR "Treatment Outcome"[Mesh]) OR "Comparative Effectiveness Research"[Mesh]) OR "Prognosis"[Mesh]) OR "Intraoperative Complications"[Mesh]) OR "Postoperative Complications"[Mesh]) OR "Vasospasm, Intracranial"[Mesh]) OR "Ischemia"[Mesh]) OR "Neurologic Manifestations"[Mesh]) OR "Infarction"[Mesh]) OR "Stroke"[Mesh]) OR "Hydrocephalus"[Mesh]) OR "Therapeutic Occlusion"[Mesh]) OR "Length of Stay"[Mesh]
#7	Search (("Embolization, Therapeutic"[Mesh]) OR "Endovascular Procedures"[Mesh]) OR "Prostheses and Implants"[Mesh]
#8	Search ((("Craniotomy"[Mesh]) OR "Microsurgery"[Mesh]))
#9	Search ((((((((((("Randomized Controlled"[Title/Abstract]) OR "Randomized Clinical"[Title/Abstract]) OR "Controlled Clinical"[Title/Abstract]) OR RCT[Title/Abstract]) OR Cohort[Title/Abstract]) OR "Concurrent Study"[Title/Abstract]) OR "Concurrent Studies"[Title/Abstract]) OR Prospect*[Title/Abstract])

#10	Search ((((("Randomized Controlled Trials as Topic"[Mesh]) OR "Cohort Studies"[Mesh]) OR "Prospective Studies"[Mesh])	
#11	(#1 OR #7) AND (#2 OR #8) AND (#3 OR #6) AND (#4 OR #5) AND (#9 OR #10)	1480
	the China National Knowledge Infrastructure	
#1	(SU = ('动脉瘤'+ '出血') AND SU = ('颅内' + '脑' + '基底'+ '囊状' + '交通' + '循环' +'颅底' + '蛛网膜下腔')) AND SU =('开颅'+ '夹闭' + '显微') AND SU = ('弹簧圈' + '栓塞' + '线圈' + '血管内' + '介入' + '神经放射学' + 'Guglielmi' + '假体' + '植入物') AND SU = ('死亡' + '发病' + '依赖' + '残' + '结果' + '疗效' + '效果' + '预后' + 'GOS' + 'mrs' + 'Rankin' + '并发症' + '血管痉挛' + '缺血' + '神经功能缺损' + '梗塞' + '中风' + '颅内感染' + '脑积水' + '再出血' + '消除' + '闭塞' + '再通' + '复发' + '住院时间') AND SU=('随机对照'+'队列'+'前瞻性'+'同时性研究'+)	12
	Wanfang	
#1	主题:(("动脉瘤"+"出血")*("颅内"+"脑"+"基底"+"囊状"+"交通"+"循环"+"颅底"+"蛛网膜下腔"))*("开颅"+"夹闭"+"显微")*("弹簧圈"+"栓塞"+"线圈"+"血管内"+"介入"+"神经放射学"+"Guglielmi"+"假体"+"植入物")*("死亡"+"发病"+"依赖"+"残"+"结果"+"疗效"+"效果"+"预后"+"GOS"+"mrs"+"Rankin"+"并发症"+"血管痉挛"+"缺血"+"神经功能缺损"+"梗塞"+"中风"+"颅内感染"+"脑积水"+"再出血"+"消除"+"闭塞"+"再通"+"复发"+"住院时间")*（"随机对照"+"队列"+"前瞻性"+"同时性研究")*Date:-2019 限制：期刊论文、学位论文、会议论文	71
	Sinomed	
#1	(("动脉瘤"[常用字段] OR "出血"[常用字段]) AND ("颅"[常用字段] OR "脑"[常用字段] OR "基底"[常用字段] OR "囊状"[常用字段] OR "交通"[常用字段] OR "循环"[常用字段] OR "蛛网膜下腔"[常用字段])) AND ("开颅"[常用字段] OR "夹闭"[常用字段] OR "显微"[常用字段]) AND ("弹簧圈"[常用字段] OR "栓塞"[常用字段] OR "线圈"[常用字段] OR "血管内"[常用字段] OR "介入"[常用字段]) AND ("随机对照"[常用字段] OR "队列"[常用字段] OR "前瞻性"[常用字段])	153
#2	("颅内动脉瘤"[不加权:扩展] OR "蛛网膜下腔出血"[不加权:扩展] OR "颅内出血"[不加权:扩展] OR "脑出血"[不加权:扩展]) AND ("显微外科手术"[不加权:扩展]) AND ("栓塞, 治疗性"[不加权:扩展] OR "假体和植入物"[不加权:扩展]) AND ("随机对照试验(主题)"[不加权:扩展] OR "队列研究"[不加权:扩展] OR "前瞻性研究"[不加权:扩展] ）	0

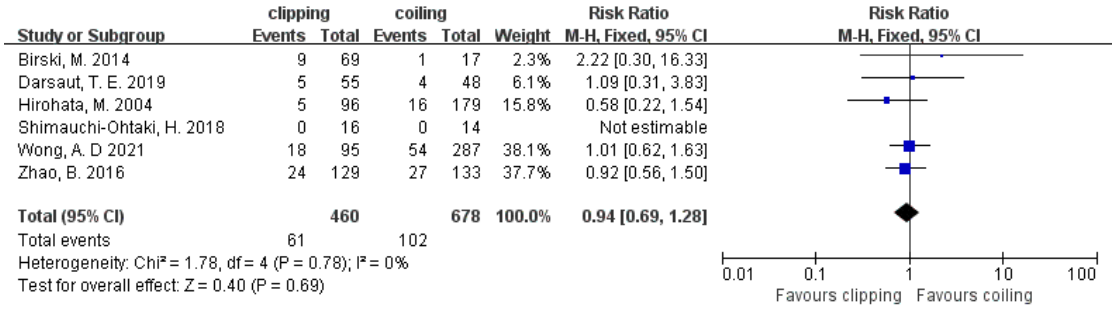


Supplemental Figure 1. The overall methodological quality of included RCTs.



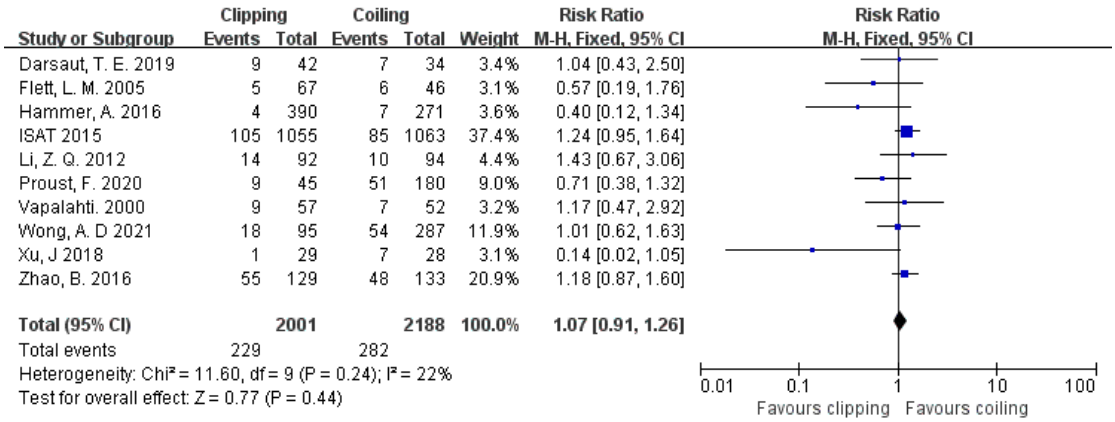
Supplemental Figure 2. Effect of clipping versus coiling on the poor outcome rate at discharge.

CI- confidence interval; M-H- Mantel– Haenszel method.



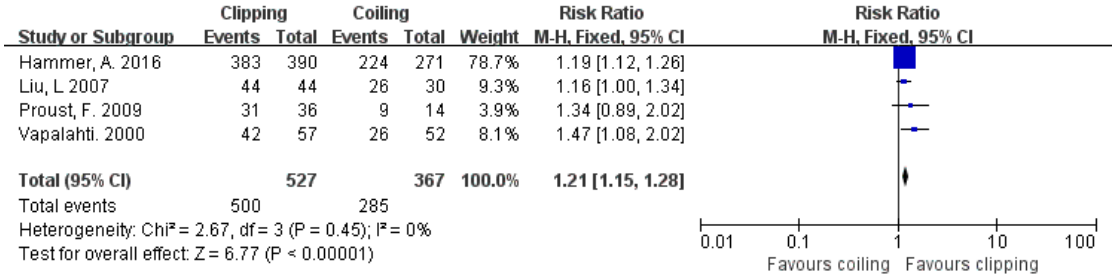
Supplemental Figure 3. Effect of clipping versus coiling on the mortality at discharge.

CI- confidence interval; M-H- Mantel– Haenszel method.



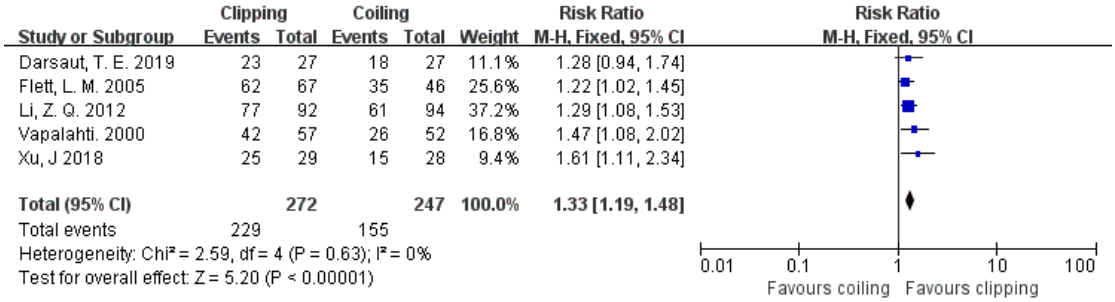
Supplemental Figure 4. Effect of clipping versus coiling on the mortality at 1-year follow-up.

ISAT- the International Subarachnoid Aneurysm Trial; CI- confidence interval; M-H- Mantel–Haenszel method.



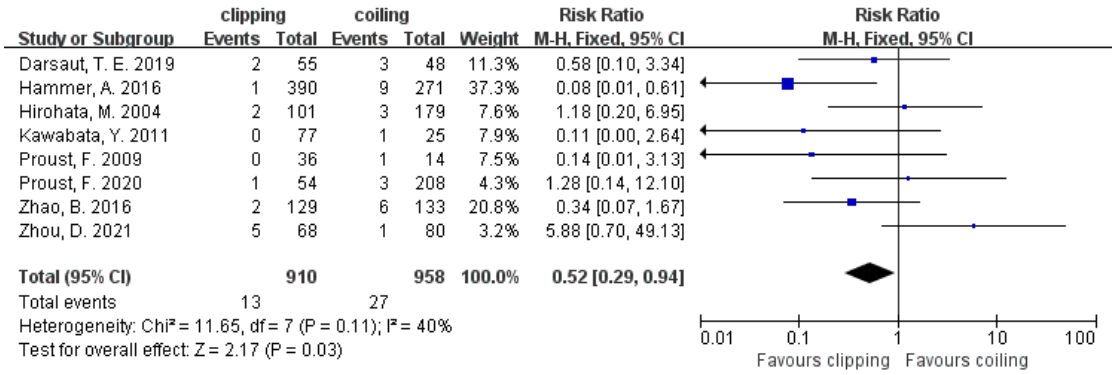
Supplemental Figure 5. Effect of clipping versus coiling on the complete aneurysm obliteration rate at discharge.

CI- confidence interval; M-H- Mantel– Haenszel method.



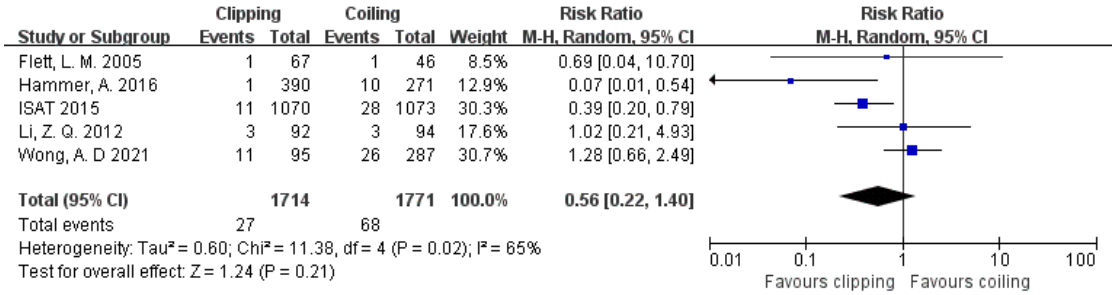
Supplemental Figure 6. Effect of clipping versus coiling on the complete aneurysm obliteration rate at 1-year follow-up.

CI- confidence interval; M-H- Mantel– Haenszel method.



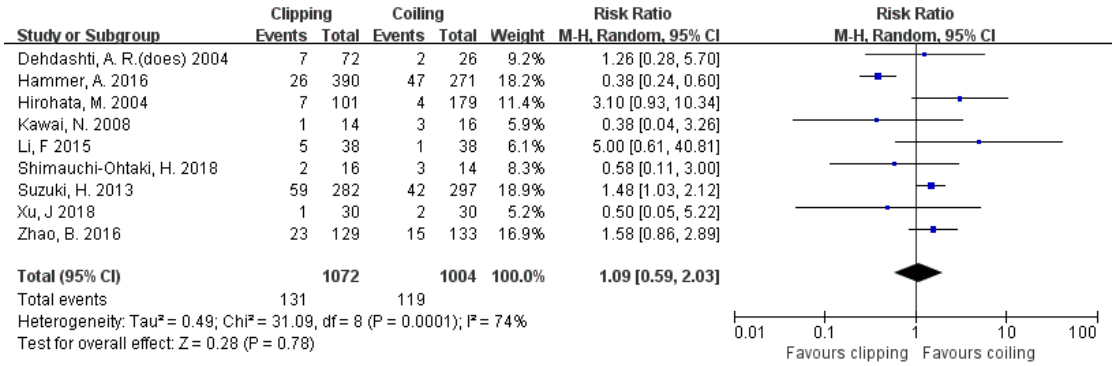
Supplemental Figure 7. Effect of clipping versus coiling on the rebleeding rate at discharge.

CI- confidence interval; M-H- Mantel– Haenszel method.



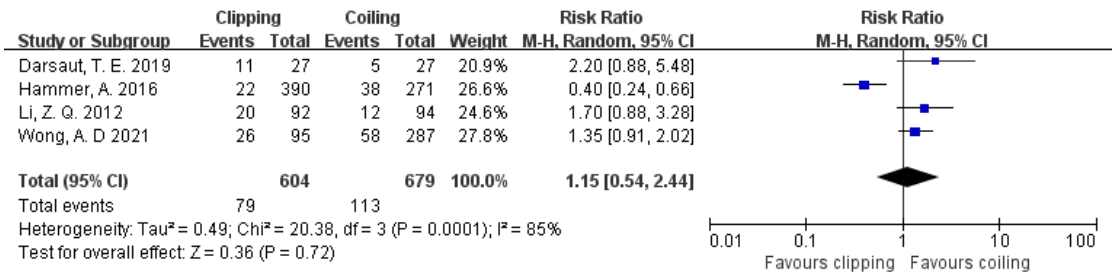
Supplemental Figure 8. Effect of clipping versus coiling on the rebleeding rate at 1-year follow-up.

ISAT- the International Subarachnoid Aneurysm Trial; CI- confidence interval; M-H- Mantel–Haenszel method.



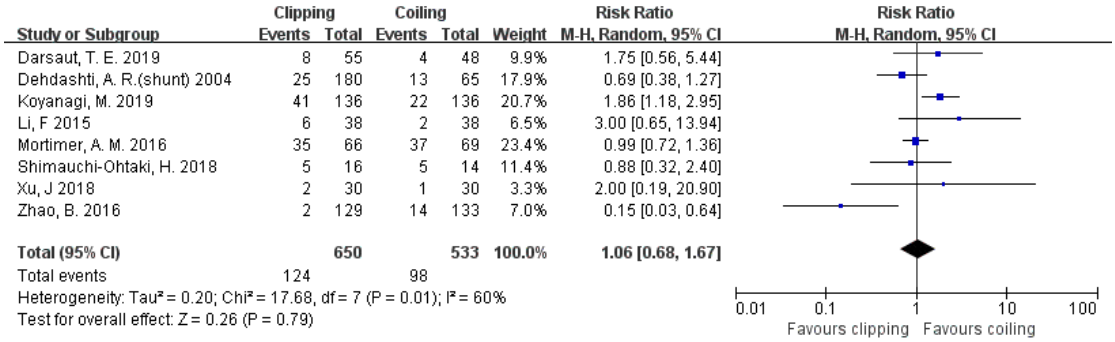
Supplemental Figure 9. Effect of clipping versus coiling on the ischemic cerebral infarction rate at discharge.

CI- confidence interval; M-H- Mantel– Haenszel method.



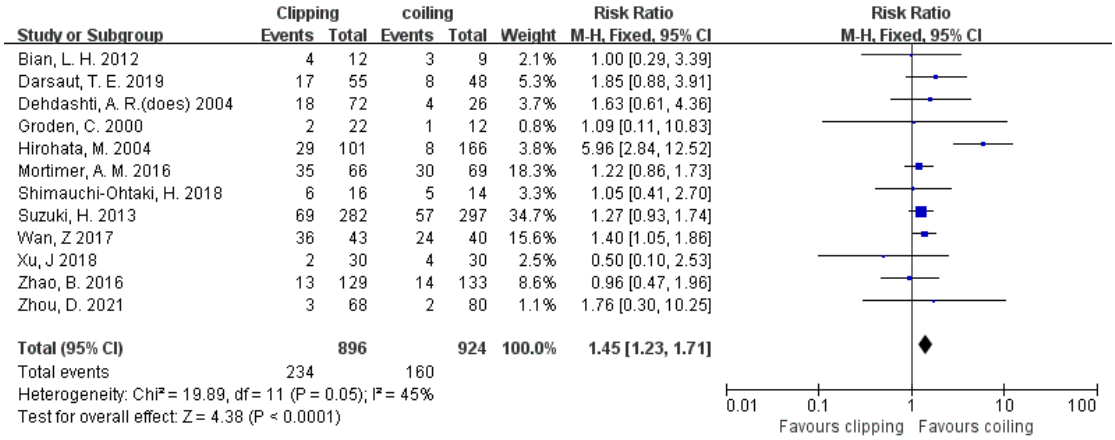
Supplemental Figure 10. Effect of clipping versus coiling on the ischemic cerebral infarction rate at 1-year follow-up.

CI- confidence interval; M-H- Mantel– Haenszel method.



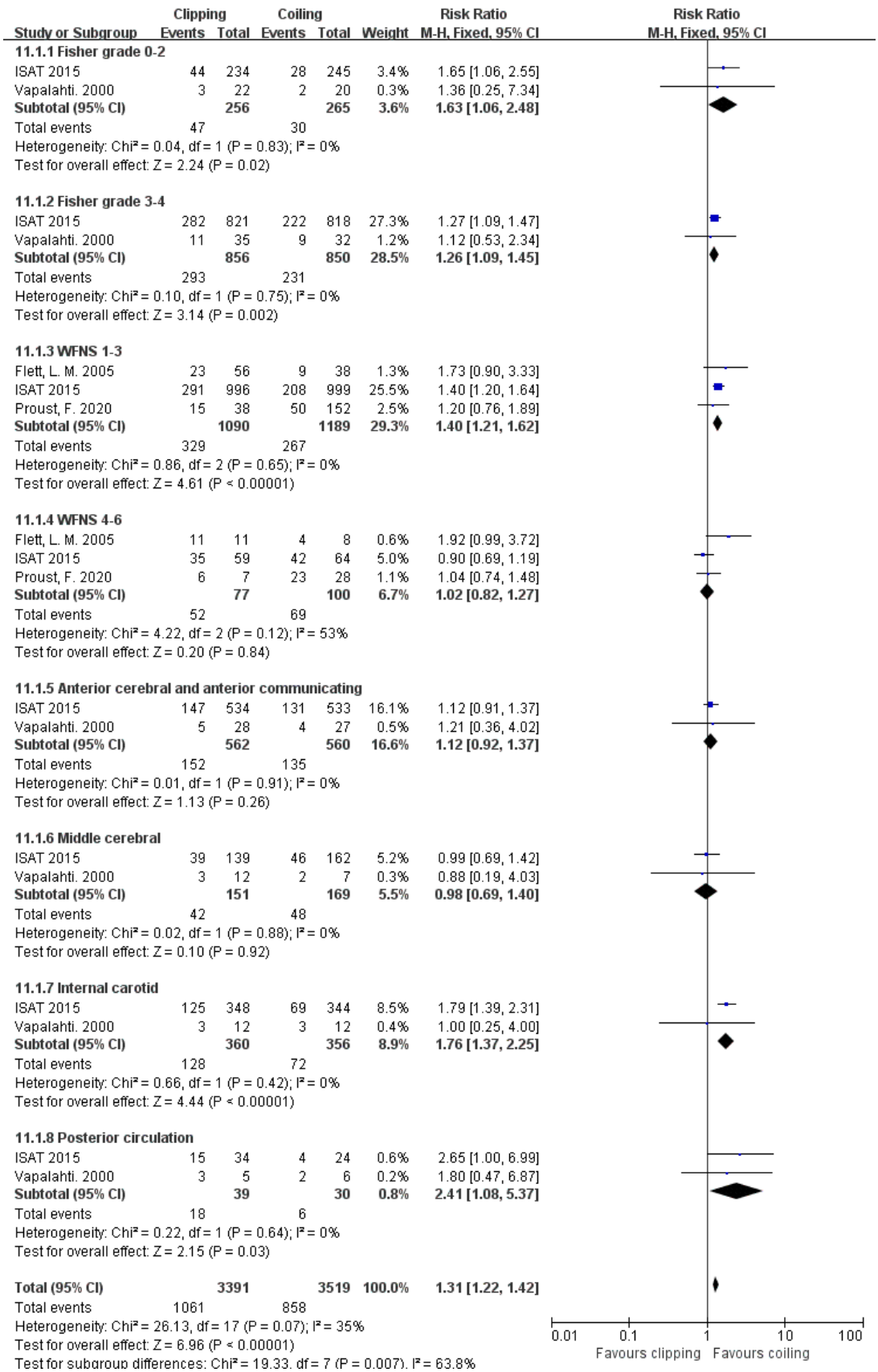
Supplemental Figure 11. Effect of clipping versus coiling on the shunt-dependent hydrocephalus rate at discharge.

CI- confidence interval; M-H- Mantel– Haenszel method.



Supplemental Figure 12. Effect of clipping versus coiling on the vasospasm rate at discharge.

CI- confidence interval; M-H- Mantel– Haenszel method.



Supplemental Figure 13: The subgroup analyses on the 1-year poor outcome rate.

ISAT- the International Subarachnoid Aneurysm Trial; CI- confidence interval; M-H- Mantel–Haenszel method.